

Concept Note: NFC-Based Hostel Access & Occupancy System

"Your Room. Verified."

Addressing Hostel Overpopulation and Squatting at the University of Lagos

1. Problem Statement

The University of Lagos operates hostels whose bed capacity was designed for a smaller student population than the university currently serves. Overcrowding is one of the most consistently cited operational frustrations among students, and a significant driver of it is squatting: students residing in rooms they were never officially allocated, occupying bed space designated for someone else.

Hostel allocation at UNILAG is currently enforced only on paper, through allocation lists managed by hostel administration. There is no physical checkpoint that verifies whether the person occupying a bed space is the person who was actually assigned to it. This gap allows squatting to persist unchecked, and it means that:

- Legitimately allocated students are frequently overcrowded out of the space they were assigned, sharing rooms, bathrooms, and utilities with more occupants than the room was built for.
- Hostel management has no reliable, real-time way to know actual occupancy per block, only the official allocation figures, which increasingly diverge from reality.
- Shared infrastructure, water supply, electrical load, and security, is strained well beyond its designed capacity, with no data trail identifying which blocks are most affected or why.
- There is no mechanism to flag or act on unauthorized occupancy in real time; issues are typically only discovered after complaints escalate.

2. Proposed Solution

We propose an NFC-based hostel access control system installed at each hostel block entrance, tying physical entry directly to verified room allocation and converting a paper-only policy into an enforceable physical checkpoint.

How it works:

- Each officially allocated student is issued an NFC tag linked to their student record and assigned room, provisioned directly against the hostel's official allocation data.
- NFC readers installed at each hostel block entrance grant access only to tags with a valid, current allocation for that specific block; unallocated or mismatched tags are denied entry.
- Both entry and exit are logged, not just entry. This is essential for computing true real-time occupancy per block; without exit logging, a student who leaves for the day still counts as "present."
- Anomalies are flagged automatically, including tags used at a hostel block that does not match the registered allocation (a student sleeping in a block they weren't assigned to), and tags left unused for an extended period, which flags a bed as a likely no-show and a candidate for reallocation to waitlisted students.

- A lightweight web dashboard (FastAPI, PostgreSQL/Supabase, Redis for live state) presents administrators with real-time occupancy versus official capacity per block, plus a no-show/reclaimable-bed view.
- A guest/visitor mode issues short-validity access for legitimate visitors, so the system does not unintentionally block people with valid reasons to be on-site.

For the hackathon demonstration, one or two physical NFC readers at hostel entrances are sufficient to prove the core mechanism, with the rest of the hostel network simulated using representative data.

3. Campus Impact

- **Restores fair access to allocated space.** Students who were properly allocated a room regain the living conditions they were promised, rather than competing for space with unauthorized occupants.
- **Gives administration verifiable data, not guesses.** Hostel management moves from estimating overpopulation to seeing exact, real-time occupancy figures per block, which can inform future hostel planning, maintenance scheduling, and capacity decisions.
- **Reduces strain on shared utilities.** Water, electricity, and security systems designed for official capacity are less likely to be overwhelmed once unauthorized occupancy is curtailed.
- **Improves campus security.** Tying entry to verified identity reduces the ambiguity around who is actually resident in a hostel block at any given time.
- **Creates a reusable data asset.** Historical occupancy and access logs give the university an evidence base for future decisions, such as whether additional hostel capacity is genuinely needed or whether enforcement alone resolves much of the reported overcrowding.

4. Research Depth

Hostel scarcity is widely and consistently reported as the most cited operational frustration among UNILAG students, alongside administrative bottlenecks in registration and result processing, both symptoms of infrastructure that has not scaled proportionally with student population growth. Squatting is a widely acknowledged, though poorly quantified, consequence of this gap: because allocation enforcement is manual and paper-based, there is currently no authoritative, real-time figure for how many hostel occupants are unauthorized at any given time.

This concept draws on established patterns from access-control and IoT-based occupancy systems used in other resource-constrained institutional settings, where low-cost NFC or RFID checkpoints have been used to convert informal or paper-based access policies into enforceable, auditable systems without requiring full infrastructure replacement. The approach is intentionally scoped to be pilotable: a small number of physical entry points can validate the core allocation-verification and anomaly-detection logic before any discussion of campus-wide rollout.

Further research required before full deployment includes: obtaining accurate, current hostel allocation data from university administration to seed the system correctly; assessing power and network reliability at proposed installation points, given known campus infrastructure constraints; and evaluating privacy and data-handling practices for storing student identity and movement logs,

in line with the Nigeria Data Protection Regulation (NDPR).